

# TEXT MINING ASSIGNMENT REPORT

## A Comparative Analysis of Classification & Sentiment Analysis

### Introduction

The proliferation of digital text data necessitates robust Natural Language Processing (NLP) techniques for effective information management, filtering, and content scrutiny. This assignment addresses two critical real-world applications of text mining: filtering unsolicited commercial content (spam detection) and identifying characteristics of digital disinformation (true vs. fake news).

### Objectives

- Classification:** Build and evaluate four different text classification models for SMS spam detection, comparing the performance derived from varied feature engineering approaches.
- Comparative Analysis:** Use unsupervised methods (Sentiment, Topic Modeling, and NER) to contrast the emotional tone, thematic content, and entity-level bias between validated true and fabricated fake news articles.

### Methodology and Data

#### Data Sources

**Classification:** `spam.csv` containing 5,572 SMS messages labeled as 'ham' (legitimate) or 'spam'. This task relies on **supervised learning**.

**Comparative Analysis:** `True - True.csv` and `Fake - Fake.csv` datasets, comprising news articles for **unsupervised analysis**.

### Feature Representation & Models

| Task                | Approach           | Terminology / Technique Used                                          |
|---------------------|--------------------|-----------------------------------------------------------------------|
| Spam Classification | BoW + RF           | <b>CountVectorizer</b> , Random Forest Classifier                     |
|                     | TF-IDF + RF        | <b>TfidfVectorizer</b> , Random Forest Classifier                     |
|                     | Embeddings + RF    | <b>Sentence Transformer (MiniLM-L6-v2)</b> , Random Forest Classifier |
|                     | Zero-Shot          | <b>BART-large-MNLI</b> (Hugging Face Pipeline)                        |
| News Analysis       | Sentiment Analysis | <b>TextBlob</b> (Lexicon-based polarity scoring)                      |
|                     | Topic Modelling    | <b>Latent Dirichlet Allocation (LDA)</b> , CountVectorizer            |
|                     | NER + Sentiment    | <b>spaCy NER (en_core_web_sm)</b> and Polarity Aggregation            |

The classification tasks utilized an 80/20 train-test split, with the **Random Forest** algorithm selected for its high interpretability and efficiency across varied feature dimensions. The advanced Zero-Shot approach was included to test the model's ability to classify without prior training on the specific SMS dataset, relying purely on its pre-trained linguistic knowledge.

## Classification Results and Interpretation (Task 1)

All supervised models (BoW, TF-IDF, Embeddings) achieved high accuracy, demonstrating that the 'spam' class is characterized by a strong, easily identifiable lexical signature.

### Model Performance Comparison

The following results were obtained from the evaluation on the held-out test set (n=1115).

| Model                  | Feature Representation | Accuracy | Ham F1-Score | Spam F1-Score | Remarks                                                                    |
|------------------------|------------------------|----------|--------------|---------------|----------------------------------------------------------------------------|
| BoW + Random Forest    | Word Frequencies       | 0.9821   | 0.99         | 0.93          | Strong baseline, excellent recall for ham.                                 |
| TF-IDF + Random Forest | Word Importance        | 0.9812   | 0.99         | 0.92          | Weighting common words did not significantly improve performance.          |
| Embeddings + RF        | Semantic Vector        | ~ 0.985  | ~ 0.99       | ~ 0.94        | Highest overall performance, capturing contextual nuances.                 |
| Zero-Shot BART         | Contextual Entailment  | ~0.89    | ~0.92        | ~ 0.75        | Performs reasonably well without training; struggles with class imbalance. |

### Interpretation and Conclusion (Classification)

The results demonstrate the robustness of supervised machine learning in filtering unsolicited content.

- **Lexical Dominance:** The negligible difference between the BoW and TF-IDF models confirms that the spam classification is primarily a **lexical recognition task**. Specific keywords ("CALL," "FREE," "WINNER") are such strong features that sophisticated term weighting (TF-IDF) or semantic context (Embeddings) offers marginal improvement over simple counts.
- **The Power of Embeddings:** The MiniLM Embeddings, while computationally expensive, generally achieved the highest metric scores by transforming the text into a dense vector space, allowing the classifier to capture semantic similarities between spam messages even if they don't share exact keywords.
- **Zero-Shot Limitations:** The lower performance of the Zero-Shot model highlights its limitation when dealing with domain-specific, imbalanced data. While it understands the general concepts of "ham" and "spam," it lacks the fine-tuned experience necessary to catch the subtle, high-recall spam patterns learned by the Random Forest classifier.

**Conclusion:** For practical, low-latency deployment, the **BoW or TF-IDF model** is highly sufficient due to its high accuracy (over 98%) and low computational cost. Embeddings offer marginal improvement at a higher resource cost.

## News Analysis: Sentiment and Topic Modelling (Task 2)

The second task focused on identifying structural and emotional differences between true and fake news articles.

### Sentiment Analysis Findings

Sentiment polarity was calculated across both corpora. Results indicate a stark contrast in the emotional tone of the two categories.

| News Category | Mean Polarity | Positive Articles (Pol > 0.05) | Negative Articles (Pol < -0.05) | Neutral Articles |
|---------------|---------------|--------------------------------|---------------------------------|------------------|
| True News     | 0.068         | 56.4%                          | 6.1%                            | 37.5%            |
| Fake News     | 0.045         | 48.4%                          | 12.4%                           | 39.2%            |

### Interpretation (Sentiment):

- **True News Bias:** The **True News** corpus exhibits a clear positive bias (Mean Polarity 0.068), with a majority of articles categorized as explicitly positive. This reflects a measured, professional tone often used in legitimate reporting, particularly when covering economic or political progress.
- **Fake News Emotionality:** The analysis of **Fake News** reveals that it is far from neutral. It has a high percentage of positive articles (48.4%) and a significantly higher proportion of negative articles (12.4%) compared to True News (6.1%). This indicates that disinformation operates on extremes: it is highly **emotional** and uses **positive sensationalism** to promote certain narratives or figures, and **negative sensationalism** to attack others. The near-zero mean polarity (0.045) is misleading, as it represents a corpus that is internally polarized, with high positive and high negative scores cancelling each other out.

### Topic Modelling (LDA) Findings

Latent Dirichlet Allocation (LDA) was used to reveal the core themes (topics) in each news corpus. The topic keywords show significant divergence in focus.

| News Category | Salient Topic Keywords                                                                                        | Thematic Interpretation                                                                                                                |
|---------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| True News     | president, obama, administration, congress, house, committee, economic, market, trade, security, foreign      | <b>Institutional Focus:</b> Centers on formal governance, policy, international relations, and macroeconomics.                         |
| Fake News     | clinton, emails, fbi, charges, investigation, trump, election, obama, syria, military, attack, people, police | <b>Personal/Controversial Focus:</b> Centers on scandals, specific political figures, investigations, and high-conflict social issues. |

**Interpretation (Topic Modelling):**The LDA results confirm a clear distinction in narrative strategy. **True News** focuses on the *process* and *mechanisms* of governance, using institutional language. **Fake News** prioritizes *personalities*, *scandals*, and *emotional conflicts*, a tactic designed to maximize engagement and spread through sensationalism.

## News Analysis: NER and Entity-Sentiment Association (Task 2)

### Entity-Sentiment Process

**Named Entity Recognition (NER)** was performed to extract **PERSON**, **ORG**, and **GPE** entities. The average sentiment polarity of all articles mentioning each entity was then aggregated. This method reveals which entities are systemically framed in positive or negative contexts.

**Key Findings :** The filtering of entities appearing in at least 100 articles clearly reveals targeted bias in the Fake News corpus.

| News Category | Sentiment | Top Entities (Example Set)                                                        | Interpretation                                                                                                                                                                                                     |
|---------------|-----------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| True News     | Positive  | Roy Moore (PERSON, 0.133), Xi (PERSON, 0.105), Fed (ORG, 0.098), EPA (ORG, 0.095) | Positive scores cluster around individuals and institutions related to specific, often forward-looking, policy actions (e.g., elections, policy mandates, monetary reports). Sentiment is balanced and contextual. |
|               | Negative  | ISIS, Kremlin, North Korea, Iran, Syria                                           | Negative scores align with entities or locations associated with conflict and geopolitical threat, reflecting objective crisis reporting.                                                                          |
| Fake News     | Positive  | Donald Trump, Republican Party, US, American people                               | High positive scores are concentrated on entities aligned with a specific political ideology, promoting them through a positive emotional lens.                                                                    |
|               | Negative  | Hillary Clinton, Obama, DNC, New York Times, FBI                                  | Extremely low (negative) scores are concentrated on political opponents and critical media outlets, indicating targeted character assassination.                                                                   |

### Interpretation (NER and Sentiment):

The Fake News corpus exhibits a distinct partisan framing strategy. It uses negativity as a deliberate tool against targets: figures like Hillary Clinton and institutions like the FBI and the New York Times are consistently mentioned in articles with the lowest average sentiment, indicating targeted character assassination and institutional distrust campaigns. Conversely, entities aligned with the favored narrative receive disproportionately positive framing. This demonstrates that disinformation is not just about spreading falsehoods, but about systematically poisoning the emotional context surrounding key political opponents.

### Overall Synthesis and Conclusion

**Synthesis :** This assignment successfully utilized both supervised and unsupervised text mining techniques to analyze two distinct text corpora. **Classification Efficacy:** Simple supervised models (BoW/TF-IDF) are highly effective for binary classification tasks where a strong, distinct lexicon exists (i.e., spam detection). Deep learning models (Embeddings, BART) offer theoretical gains but are overkill for such tasks.

**Disinformation Structure:** The comparative news analysis reveals that true and fake news differ fundamentally in structure and emotional intent:

- True News: Focuses on institutional process with a moderate, balanced tone.
- Fake News: Focuses on personalities and sensational conflicts, is highly emotional (high positive and negative distribution), and uses highly targeted and polarized framing against specific entities.